0012 — The OR12 Base-Rate Card: Replicating the IB Statistics, and What They're Really Made Of — 2026-07-07

Series: MC Setup Research Notes · Note 0012 **Confidence:** High — 1,251 ES days (2021–2026) of our own tick-built 5M data, simple counts (no fitting), era-split stability shown per cell, and the two headline third-party claims stress-tested with controls. No trading claim; no cost model applies. **TL;DR:** We replicated the published Initial-Balance statistics (note 0011 §4.1) on our own data — and then dissected them. Three findings: (1) the celebrated **"IB extreme formation order" skew is mostly a proximity artifact** — the real conditioner is where price sits within the IB at 10:30 (low third → ~85% first break down AND 2.5:1 odds the day closes below the IB; mirrored at the high); (2) **"narrow IB → range extension" inverts in ADR units** — wide IBs are followed by bigger afternoons (post-IB range >0.5×ADR: ~60% narrow vs ~90–99% wide); IB width is a realized-volatility nowcast that predicts afternoon range, not direction; (3) trend days concentrate on **wide**-IB days (38–41% vs 9–12% narrow), the opposite of the folklore reading. The card (bucket × IB-width × 10:30 location) is now validated on our data and ready to be the foundation of the 9:35 context screen.

1. The setup (so this note stands alone)

- **Data:** 1,251 ES RTH sessions (2021-07 → 2026-07), 5-minute bars built from tick data, back-adjusted continuous.
- **IB** = first 12 bars (08:30–09:30 CT). All conditioning features are known by 09:30 CT (open bucket, gap) or at the IB close (IB width vs ADR14, formation order, bar-12 close location). Outcomes are end-of-day.
- **Why:** note 0010's verdict was to build a context display, whose foundation is honest conditional base rates. Note 0011 catalogued third-party ES numbers (TradingStats, 2,686 days 2015–25); nothing goes on a live screen until replicated here — this note is that replication, plus the tautology controls the third-party source never ran.
- **Definitions:** bucket = open location vs yesterday's range (above_yH / upper_half / lower_half / below_yL); IB-width tiers = full-sample terciles of IBrange/ADR14 (cuts: 0.393, 0.582); trend day = RTH range > 1.1×ADR20 AND close within 25% of the day's extreme; extension = trading >0.5 IB-widths beyond either IB extreme after 09:30.

2. The question

Do the documented IB conditioners (width-vs-ATR, extreme formation order) replicate on our 2021–26 data — and are they real information or measurement artifacts?

3. How we tested it

1. Main card: bucket × IB-width tier → P(trend), P(extension, IB units), P(post-IB range >0.5×ADR), median follow-through in ADR, P(both sides broken), P(day close above/inside/below IB), with a 2021–23 vs 2024–26 era split per cell. 2. Formation-order replication: which IB extreme formed first → first break side. 3. **Proximity control:** the same, within bands of where bar 12 closes inside the IB (low/mid/high third). 4. **Denominator control:** extension measured both in IB multiples (their convention) and in ADR units (ours). 5. Bar-12 close location → day close vs IB (position persistence vs directional prediction).

4. Results

4.1 The main card (bucket × IB-width tier), key columns

bucket	tier	n	P_trend	P_ext (IB units)	P(postIB rng>0.5 ADR)	med follow-thru (ADR)
above_yH	narrow	111	9.9%	90.1%	56.8%	0.23
above_yH	mid	94	16.0%	72.3%	83.0%	0.29
above_yH	wide	73	38.4%	56.2%	89.0%	0.35
below_yL	narrow	34	5.9%	73.5%	67.6%	0.18
below_yL	mid	77	23.4%	67.5%	85.7%	0.36
below_yL	wide	93	33.3%	47.3%	98.9%	0.39
lower_half	narrow	101	11.9%	81.2%	68.3%	0.28
lower_half	mid	112	25.0%	75.9%	89.3%	0.38
lower_half	wide	116	41.4%	53.4%	95.7%	0.38
upper_half	narrow	171	8.8%	81.3%	60.2%	0.23
upper_half	mid	134	23.9%	73.9%	82.8%	0.36
upper_half	wide	135	39.3%	57.8%	91.9%	0.46

Unconditional: trend 23%, extension 70%, both-sides broken 27%. Era split (2021–23 vs 2024–26): trend rates stable within ~±7pp for all cells with n>90 except upper_half/narrow (15.6% → 4.7% — flag). Full card incl. close-vs-IB columns: docs/living/or12_baserate_tables_20260707.csv.

Read: the two extension columns tell opposite stories, and both are partly mechanical (§5). The robust, non-tautological gradients: **trend days concentrate on wide IBs** (9–12% narrow → 33–41% wide, stable across eras) and **afternoon range grows with IB width** in ADR units.

4.2 Formation order replicates… (raw)

IB high formed first → first break down 69.0% / up 28.0% (n=607); low first → up 73.4% / down 25.3% (n=644). Stronger than the documented 45/24 — suspiciously strong.

4.3 …and mostly dies under the proximity control

First-break-down rate, by formation order WITHIN bar-12 close-location bands:

bar-12 closes in	hi-first: down%	lo-first: down%	formation-order residual
low third of IB	86.2 (n=363)	80.0 (n=45)	+6pp
mid third	50.3 (n=179)	45.9 (n=148)	+4pp
high third	24.6 (n=65)	13.1 (n=451)	+11pp

The real variable is where price sits at 10:30 (low third → ~85% first break down regardless of order); formation order contributes a +4…+11pp residual. The documented "2:1 skew" is mostly proximity in disguise. Corollary: the n-splits show formation order and location are heavily entangled (high-formed-first days usually are sitting low at 10:30).

4.4 Bar-12 location also predicts the day CLOSE — position persists, direction doesn't

bar-12 closes in	day closes above IB	inside	below IB	P_trend	n
low third	17.6%	38.5%	43.9%	27.9%	408
mid third	37.6%	36.7%	25.7%	19.3%	327
high third	48.8%	32.9%	18.2%	22.5%	516

~2.6:1 odds the day closes on the side price already occupies at 10:30. Crucial nuance: this is **position persistence, not a forward-return signal** — the sign of the move from the 10:30 price remains ~50/50 (note 0010 test [2]). Both can be true at once: price that is low tends to stay low without necessarily going lower.

4.5 The denominator dissection (why the folklore inverts)

- "Narrow IB → 90% extension" (IB units) is favored mechanically: half an IB-width is a small absolute move when the IB is narrow.
- In ADR units the ranking flips: post-IB range >0.5×ADR on ~60% of narrow-IB days vs ~90–99% of wide-IB days; median follow-through 0.18–0.28 ADR narrow vs 0.35–0.46 wide.
- Interpretation: **IB width vs ADR14 is a same-day realized-volatility nowcast** (ADR14 lags; a wide IB means vol has expanded today, and intraday vol clusters). It predicts how far the afternoon travels — in either direction — and hence trend-day odds. It says nothing about side.

5. Why it works / fails (the synthesis)

The third-party numbers replicate, but two of the three headline effects are partly measurement geometry: formation order proxies for 10:30 location; IB-unit extension proxies for the denominator. What survives dissection as genuinely informative at 10:30 is a clean three-factor card: **(1) open-location bucket** (auction context vs yesterday), **(2) IB width vs ADR** (vol nowcast → afternoon range & trend-day odds), **(3) bar-12 close location within IB** (position persistence → which side the day likely settles on + first-break side). All three are already features in the OR12 fingerprint — the card is the interpretable 3-axis projection of what the twin-matcher uses in 80 dimensions.

6. Recommendation

Adopt the 3-factor card (bucket × IB-width tier × 10:30 location) as the foundation of the 9:35 context screen. Display real per-cell rates with n; grey out cells with n<50; show the upper_half/narrow era instability flag. Drop formation order from the display (keep as a fingerprint feature) — it's redundant with location and would double-count. Frame the location factor as "the day tends to settle on this side," never as "price will go this way from here." No trade recommendation is made or implied.

7. Caveats & open questions

- Cell n runs 34–171; below_yL/narrow (n=34) is unusable — grey it out.
- Terciles are full-sample cuts (mild look-ahead for a historical card); the live screen should use trailing cuts — trivial change, queued with the screen build.
- Trend-day thresholds (1.1×ADR, 25% band) inherited from 0011's community definitions, sanity-checked but not optimized (deliberately — optimizing labels invites circularity).
- upper_half/narrow trend rate collapsed in 2024–26 (15.6% → 4.7%) — watch item.
- C-period confirmation (10:30–11:00 close outside IB) not yet replicated — natural v2 column for the card.

8. Reproduce

- scripts/or12_baserate_tables.py (one run, ~40s) → tables above + docs/living/or12_baserate_tables_20260707.csv
- Outcome extractor shared with 0010: scripts/or12_outcome_agreement.py (day_outcome, incl. first_break, post_rng_pts, abs_ret_pts)
- Fingerprint/context builder: scripts/or12_pattern_groups.py (build_features, ib_atr, close_loc, buckets)
- Companions: note 0010 (twin-matcher verdict), note 0011 (source survey),

docs/living/or12_research_daytype_20260706.md